Project 6: Privacy Project

1. Review of Incidents

1A. Security vs. Privacy Incidents

- **PowerSchool Data Breach:** This incident qualifies as both a security and privacy incident. The breach involves unauthorized access to student records, which indicates a failure in security measures. Additionally, since personal student data is exposed, it constitutes a privacy violation.
- **Elon Musk's DOGE Website Defacement:** This is primarily a security incident. The ability for anyone to edit the website suggests a vulnerability in access controls, but there is no clear exposure of personal data, making it less of a privacy concern.
- **DC Housing Authority Surveillance Lawsuit:** This is a privacy incident. The lawsuit alleges unconstitutional surveillance, implying a violation of individuals' expectations of privacy without necessarily involving unauthorized access to data, which would make it a privacy issue.

1B. Contextual Integrity Analysis

- **PowerSchool Data Breach:** This violates contextual integrity because student records are expected to be used solely by authorized school personnel, not exposed publicly.
- **Elon Musk's DOGE Website Defacement:** Contextual integrity is not necessarily violated as there is no expectation that the website should be not changeable, though security expectations were certainly violated.
- **DC Housing Authority Surveillance:** This violates contextual integrity because residents have an expectation that their private lives are not subject to government surveillance without due process.

2. Shamir Secret Sharing

2A. Finding the Secret

We are given the points $(1, 62), (2, 91), (3, 124), (4, 161), (5, 202), (6, 247)$ and $(7, 296)$ from a polynomial $f(x)$. Using three points to reconstruct the polynomial, we find $f(0)$, the secret.

By setting up a system of equations, we can reconstruct with three ($k = 3$) points a second-degree ($k - 1$) polynomial $f(x) = ax^2 + bx + c$. Solving for coefficients with the given points,:

Step 1: Setting up the system of equations

Using three of the given points, we generate the system of equations:

1. $f(1) = a(1)^2 + b(1) + c = 62 \Rightarrow a + b + c = 62$
2. $f(2) = a(2)^2 + b(2) + c = 91 \Rightarrow 4a + 2b + c = 91$
3. $f(3) = a(3)^2 + b(3) + c = 124 \Rightarrow 9a + 3b + c = 124$

Step 2: Solving for a, b, c

We can subtract the first equation from the second:

$(4a + 2b + c) - (a + b + c) = 91 - 62 \Rightarrow 3a + b = 29$

And then subtract the second equation from the third:

$(9a + 3b + c) - (4a + 2b + c) = 124 - 91 \Rightarrow 5a + b = 33$

Now we can solve the sub-system:

$3a + b = 29$

$5a + b = 33$

Subtract the first from the second:

$(5a + b) - (3a + b) = 33 - 29$

$2a = 4$

$a = 2$

Substituting $a = 2$ into $3a + b = 29$:

$3(2) + b = 29$

$6 + b = 29$

$b = 23$

Substituting $a = 2$ and $b = 23$ into $a + b + c = 62$:

$(2) + (23) + c = 62$

$c = 37$

Thus, the secret is:

$\mathbf{f(0) = 37}$

2B. Scheme Parameters

Since any *three* points suffice to reconstruct $f(x)$, this is a **3-out-of-7** Shamir secret sharing scheme, with **k = 3** and **n = 7**.

3. Missile Launch Secret Sharing

Let us start to define a weighted Shamir secret sharing scheme for this scenario such that there are no other minimal groups:

First, we translate the minimal groups into a linear system of equations:

- 3 four-star generals:
  - Let f represent a four-star general, and s represent the secret code $\Rightarrow 3f = s$
- 2 four-star generals and 7 colonels:
  - Let c represent a colonel $\Rightarrow 2f + 7c = s$
- 1 four-star general, 2 three-star generals and four colonels:
  - Let t represent a three-star general $\Rightarrow f + 2t + 4c = s$
- Four three-star generals and one colonel:
  - $\Rightarrow 4t + c = s$

Solving the system of equations:

1. Solving for f in the first equation:
   a. $3f = s \Rightarrow f = \frac{s}{3}$
2. Substituting f into the second equation:
   a. $2(\frac{s}{3}) + 7c = s \Rightarrow 7c = \frac{s}{3} \Rightarrow c = \frac{s}{21}$
3. Substituting f and c into the third equation:
   a. $(\frac{s}{3}) + 2t + 4(\frac{s}{21}) = s \Rightarrow 2t = \frac{21s}{21} - \frac{11s}{21} \Rightarrow t = \frac{5s}{21}$
4. Then, expressing f and t in terms of c, and expressing s in terms of c:
   a. $f = 7c \;\Big|\; t = 5c \;\Big|\; 21c = s$
   b. The above implies that 21 colonels must come together to recover the secret, whereas a single three-star general has 5 times as much secret-recovering power as a colonel, and that a four-star general has 7 times as much secret-recovering power as a colonel.

As such, if we assign one share to a colonel, we then find the resulting weighted Shamir secret sharing scheme, represented as a position-share table:

Position	Shares
Four-star general	7
Three-star general	5
Colonel	1

In total, this weighted Shamir secret sharing scheme assigns 48 shares in total:

- $3_{\text{four-star generals}} \times 7_{\text{individual shares}} + 4_{\text{three-star generals}} \times 5_{\text{individual shares}} + 7_{\text{colonels}} \times 1_{\text{individual shares}} = (21 + 20 + 7) = 48 \text{ total shares}$

We can verify that the provided minimal groups all add up to the share threshold of 21 shares:

1. 3 four-star generals:
    a. $\Rightarrow 3 \times 7 = 21$
2. 2 four-star generals and 7 colonels:
    a. $\Rightarrow (2 \times 7) + (7 \times 1) = 14 + 7 = 21$
3. 1 four-star general, 2 three-star generals and four colonels:
    a. $\Rightarrow (1 \times 7) + (2 \times 5) + (4 \times 1) = 7 + 10 + 4 = 21$
4. Four three-star generals and one colonel:
    a. $\Rightarrow (4 \times 5) + (1 \times 1) = 20 + 1 = 21$

However, there are indeed some other minimal groups—in quantity, four—that are not supersets of the ones given and that also satisfy agency member quantity constraints:

1. 2 four-star generals and 2 three-star generals:
    a. $(2 \times 7) + (2 \times 5) = 24 > 21$ (enough shares!)
2. 2 four-star generals, 1 three-star general, and 2 colonels
    a. $(2 \times 7) + (1 \times 5) + (1 \times 2) = 21$ (enough shares!)
3. 1 four-star general and 3 three-star generals:
    a. $(1 \times 7) + (3 \times 5) = 22 > 21$ (enough shares!)
4. 3 three-star generals and 6 colonels:
    a. $(3 \times 5) + (1 \times 6) = 21$ (enough shares!)

We may now attempt to find some other minimal groups, but there is no possibility of any other minimal groups existing, since none of all the other possible agency member combinations we can think of can achieve at least 21 shares, the minimum amount needed for recovering the secret, due to constraints on the agency's member quantities:

1. 2 three-star generals and 11 colonels:
    a. $(2 \times 5) + (1 \times \cancel{11}) = 21$ (not enough members!)
2. 1 four-star general and 14 colonels:
    a. $(2 \times 5) + (1 \times \cancel{14}) = 21$ (not enough members!)
3. So on and so forth...

As such, the four other minimal groups derived above are the only other minimal groups that satisfy the agency's member quantity constraints as well as the share threshold constraint.

4. CFPB Narrative Scrubbing Standard Assessment

4A. Categories of Protected Information

1. Redacted Information

This method involves completely removing specific details from complaint narratives to reduce the likelihood of individuals being identified gratuitously. The categories include:

- **Names**: Full names of individuals are redacted.
- **Ages**: Specific age information is redacted.
- **Legal Representation**: Details about attorneys or legal representatives are redacted.
- **Zip Codes:** Although they are fully divulged, zip codes are only redacted when a consumer's city population is well below 20,000 people.
    - **Redaction continues into** things such as physical characteristics or personal descriptors like race, ethnicity, sexual orientation, nationality, or immigration status, contact information like addresses, phone numbers, or email addresses, financial account information, including account numbers, credit card numbers, or other financial identifiers, SSNs and other identifying numbers like driver's license numbers or passport numbers, offensive language, and etc.

2. Generalized Information

Sometimes, specific details are replaced with less specific terms so as to retain the context that a narrative is situated in, while also upholding privacy:

- **Dates**: Specific dates may be generalized to a month or year.
- **Locations**: Exact addresses might be replaced with broader geographic areas, such as zip codes, which are only generalized with cities with lower than 20,000 people in terms of population (and may even be redacted in more extreme cases).
- **Dollar amounts:** Exact dollar amounts are rounded to prevent exact amount identification, but to also preserve informational context that dollar amounts provide.

4B. Re-identification Risks

- Even after redaction, narrative details may allow linkage to public records, previous complaints, or even certain individuals. For example, the presence of a full zip code in tandem with idiosyncrasies such as writing style or social status may lead to easy re-identification of a person, especially if those traits are only generalized or easy to infer.
    - "I got car loan in XXXX balance was {$1200.00} that was owe the motor blew up. The car at garage not sure if they pick it up. Its been over XXXX that the lawyer everytime get job the garnish check. I ask to settle tge attorney said his client wanted over {$7000.00} XXXX. The lawyer has been corrupted because they go each time get employ open case up in XXXX XXXX XXXX. They said in XXXX it was {$21000.00} and now attorneys saying its {$14000.00} with all court fees and lawyer. I have not been working on Social Security and before filed was on XXXX XXXX XXXX. Can someone help me this been over twenty years. I tried get legal aid but they not accepting anyone. My income monthly {$900.00} Social Security and started working 40 hours monthly and thats XXXX XXXX monthly every two weeks {$500.00} and they trying get 21 % if that now."
        - Source: https://www.consumerfinance.gov/data-research/consumer-complaints/search/detail/12446073
    - The example above clearly illustrates the consumer's certain lack of literacy in forming coherent English sentences, hinting at a certain degree of illiteracy. Moreover, the amount of hours they work per week and salary socioeconomically profile them. Such immediately apparent signs may possibly even have further implications, such as technological ineptitude due to the numerous typos, as well as the specificity of the loan type as being car loans, which only serve to exacerbate the degree to which the consumer is re-identifiable. Coupled with their "Older American" demographic tag and the exact company they filed for a complaint against, they may be easily identified within the vicinity of the provided zip code of their locality, because though it may be difficult to pinpoint a consumer within a large city, such specific details being left in the narrative with location details could very much allow for re-identification.

4C. Data Utility Shortcomings

- Excessive redaction reduces the usefulness of narrative info by obscuring important details, such as dates, which can make it difficult to accurately track complaint trends and associate issues with specific time periods or companies.

- "To Whom It May Concern, I am filing a formal complaint regarding Wells Fargos decision to close my checking (XXXX) and savings (XXXX) accounts, effective XX/XX/XXXX. I believe this action is unjustified and has created unnecessary financial hardship for me and my family. Wells Fargo has cited the closure as a result of eight claims I filed between XX/XX/XXXX and XX/XX/XXXX. I want to emphasize that each of these claims was a legitimate dispute over transaction issues, not fraudulent or made in bad faith. While I was unable to provide sufficient supporting documentation in some cases, leading to the reversal of provisional credits, I have always acted within Wells Fargos dispute resolution guidelines. The decision to close my accounts appears to be a punitive response rather than a fair assessment of my banking history and intentions. Additionally, my accounts negative balance resulted from an unforeseen financial hardship. I have been on short-term XXXX leave since XX/XX/XXXX, which caused a temporary disruption in my income. As a result, my account lacked deposits and became overdrawn. I recently received my first XXXX payment and had planned to use these funds to correct the negative balance. However, instead of allowing me to resolve the issue, Wells Fargo chose to close my accounts without consideration for my circumstances. This decision has significantly impacted my financial stability. As a single parent, I rely on these accounts to manage my household expenses and provide for my children. Losing access to my primary banking relationship has made it even more difficult to recover financially during an already challenging time. Following the submission of my complaint to Wells Fargo, I received a call on XX/XX/XXXX, from XXXX with XXXX XXXX XXXX at Wells Fargo. Since then, I have attempted to return her call five to six times and have left a message each time, yet I have not received any response. This lack of communication has further complicated my ability to resolve this issue in a timely manner. I am requesting that Wells Fargo reconsider this decision and allow me to maintain my accounts. I have made multiple efforts to engage in dialogue with the bank, but my request for reconsideration has not been acknowledged beyond the initial call. I believe this account closure is an unfair and disproportionate response that does not take into account my history as a customer or my efforts to resolve the issue. I ask the Consumer Financial Protection Bureau to investigate this matter and ensure that Wells Fargo is held accountable for its actions, including their failure to provide appropriate customer support and follow-up. I am happy to provide any additional details or documentation as needed. Thank you for your time and attention to this matter."
- The complaint above highlights an issue with Wells Fargo closing a consumer's accounts, bringing to light the fact that excessive claims were filed. However, the suppression of exact dates, like "XX/XX/XXXX" removes critical context, such as whether the closures happened during a financial crisis or maybe even a known period of banking system issues. This could limit the ability of researchers, policymakers, and even consumers, for example, to identify whether Wells Fargo has a pattern of excessively closing accounts in specific circumstances. Also, such over-redaction of corporate details, like dates, can reduce the dataset's usefulness by obscuring patterns of misconduct or systemic issues. If Wells Fargo repeatedly engages in fraudulent practices but the times of these actions are redacted, regulators and researchers may struggle to detect and address widespread problems. While privacy protections are necessary, maintaining enough detail to track industry-wide trends and recurring issues would enhance the data's utility; even just generalizing some of the time-related data would be greatly useful.

4D. Improvement Suggestions

- As shown in previous examples, there are issues that still persist even despite the existing protocols in the CFPB's scrubbing standard. An example below helps to illustrate my idea.
    - "In early 03/15/2023, I got lay off from my job as a software enginner at TechNova Inc., which left me strugglnig to keep up with my morgage payments. My monthly income dropt from $6,245 to $1,850 in unemplyment benefits, and I reach out to Chase for mortgate assitance. Despite expain my situation and providng documntation, I was repeateadly deny forbearance options. I later discovr that other borrowers in similair situations had reciv assistance, and I belive I was treat unfairedly. Now, I am at risk of foreclousure, and my repeate attempt to negociate a managable payment plan has been ignor. I request that the Consumer Financial Protection Bureau investigat Chase's inconsitent appliction of mortgate relief polocies."
    - A sample narrative before scrubbing has been provided above. Most of the issues could definitely be fixed by CFPB's existing standard, but as illustrated by a previous example, numerous typos and writing style would continue to persist past scrubbing. As such, streamlining the grammar of a passage would not only prevent re-identification via stylometry, but also help readers understand better the context of the narrative, without needing to infer what a consumer must have meant by something. Furthermore, the scrubbing process could be improved by contextual generalization. For example, rather than removing exact income figures and dates, dollar amounts could be grouped into ranges and dates could be also grouped into small ranges. This would still allow researchers and regulators to spot patterns and trends without revealing too much personal information, and would make the data more useful while sustaining privacy.

5. Course Feedback and Privacy

5A. Accuracy of the Instructor's Packet

The professor included quotes from 3 CS majors and 3 non-CS majors who recommended the course. However, providing only the median responses was a rather erroneous decision on the professor's part, as they are a single response that do not offer visibility on every other response. In other words, they do not provide a representative sample of student feedback but are instead cherry-picks, and are a positive misrepresentation of the students' feedback.

5B. True Recommendation Rate

1. Beginning with the formula for the relation between actual rate and reported rate:

   $r = p \times c + (1-p)(1-c)$

   We get:

   $c = \frac{p+r-1}{2p-1}$

   Where:

a. p is the probability of the student recommending the course

b. r is the *reported* recommendation rate for this professor's course

c. c is the *actual* recommendation rate for this professor's course

2. $p = (1 - 0.25)$, due to the flipping of the probability of a singular student's recommendation

   $r = 0.65$, as provided in the problem's description

3. $\Rightarrow c = \frac{(1-0.25)+(0.65)-1}{2(1-0.25)-1}$

4. $\Rightarrow c = \frac{0.75+0.65-1}{2\times0.75-1}$

5. $\Rightarrow c = \frac{0.4}{0.5} = 0.8$

As we have found that $c = 0.8$, it may be concluded that the true recommendation rate for this professor's course is **80%**.